Alexandria Engineering Journal 103 (2024) 376–383

Contents lists available at ScienceDirect

Alexandria Engineering Journal

journal homepage: www.elsevier.com/locate/aej

A compact ultra-wide band fractal antenna for breast cancer detection applications

Sadasivam S [a,*], Thulasi Bai [b]

[a] Department of Electronics and Communication Engineering, KCG College of Technology, India
[b] Department of Electronics and Communication, KCG College of Technology, India

ARTICLE INFO

Keywords:
Ultra-wide band
Breast cancer
SAR

ABSTRACT

A compact ultra-wide band antenna for breast cancer detection in bio-medical applications is presented in this study. The proposed antenna is printed on 1.6 mm thick FR4 substrate with a dielectric constant of 4.6 and tangential loss of 0.02. The uniform ground plane on the substrate (PCB) has been altered by shortening its length and making minimal changes to its side edges in order to improve overall bandwidth. The suggested antenna is functioning from 3 to 10.2 GHz which covers the ultra-wide band recommended by Federal Communication Commission (FCC) over the 10 dB impedance bandwidth. Additionally, the antenna has a significant overall gain of 3.2 dBi and an overall radiation efficiency of between 89 % and 92 %. Furthermore, the antenna maintains a uniform radiation pattern across its operating band. Parametric research has been conducted on the different parameters to tune the antenna into the required operating band, and placement analysis of the antenna is then investigated by assessing specific absorption rate (SAR).

1. Introduction

Globally, breast cancer stands out as the foremost cause of cancer-related deaths among women, prevailing in both developed and developing nations [1,2]. Women face double the risk of developing breast cancer compared to men [3], contributing to its widespread occurrence. Early detection through cancer screening plays a vital role in preventing disease-related mortality and morbidity, as it has been shown to enhance the prognosis of the illness. The motivation behind this research was the potential to save 2.5 million lives between 2020 and 2040 by reducing the mortality rate by 2.5 %.

Physical examinations or diagnostic imaging techniques like mammography and ultrasound serve as means to detect abnormalities in breast tissue during the early stages of breast cancer detection [4,5]. Studies conducted between 1977 and 1998 revealed false negative rates ranging from 4 % to 34 % for mammograms. Consequently, the inherent limitations of X-ray mammography contribute to increased disease-specific morbidity and mortality due to the false sense of security it provides to affected women, leading to delayed detection and treatment [6]. Moreover, this imaging modality necessitates breast compression by the patient [7,8], and it carries the risk of exposure to ionizing radiation [9]. Research indicates that women with denser breast tissue on radiographs exhibit less sensitive findings in screening mammography [10,11]

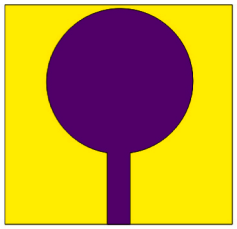
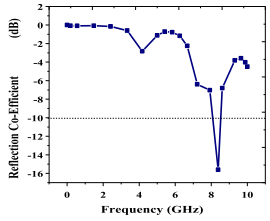
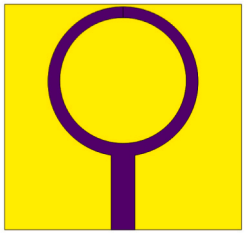
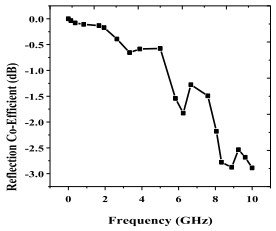
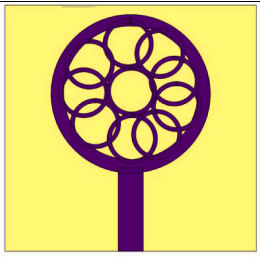
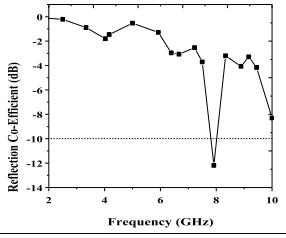
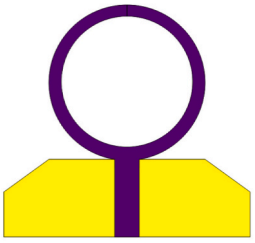
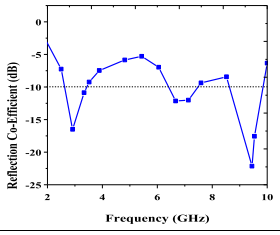
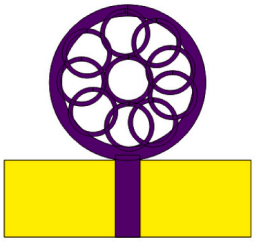
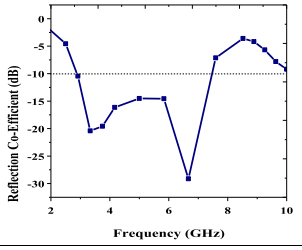
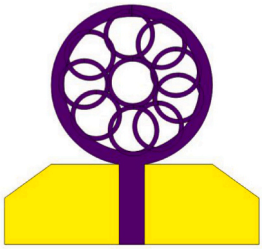
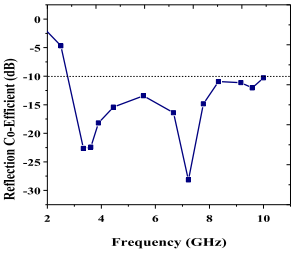
Following X-ray mammography, ultrasound imaging is commonly employed as a complementary procedure. This method assists in examining tumors that may be challenging to detect via mammography, especially in dense breast tissue. Similar to mammograms, ultrasonography serves as a cancer detection tool but has inherent limitations. Despite being a painless procedure unlike mammography, ultrasound lacks effectiveness in distinguishing between malignant and benign tumors and exhibits poor resolution [12]. Consequently, there is an immediate need to explore alternative detection methods that offer both efficiency and accuracy. Microwave imaging [13] emerges as a promising technique with diverse potential applications in biomedicine. In breast imaging, this method involves transmitting short bursts of low-power microwaves into the breast tissue through a microwave transmitter. Antennas placed around the breast collect the back-scattered radiation, and the acquired signals are utilized to generate a three-dimensional image of the scanned breast. Therefore, antennas play a crucial role in the microwave imaging breast cancer detection method.

Broad band antennas have been getting a lot of attention recently because of the new use of high frequency and multiband devices. Among

* Corresponding author.
E-mail address: sadasivam.ece@kcgcollege.com (S. S).

https://doi.org/10.1016/j.aej.2024.06.018
Received 7 November 2023; Received in revised form 29 April 2024; Accepted 6 June 2024
Available online 20 June 2024
1110-0168/© 2024 The Author(s). Published by Elsevier BV on behalf of Faculty of Engineering, Alexandria University This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Table 1
Design evolution of the proposed antenna with iterations and reflection co-efficient.

Iteration	Evolution Steps	Reflection Co-Efficient
1		
2		
3		
4		
5		
6 (Proposed)		

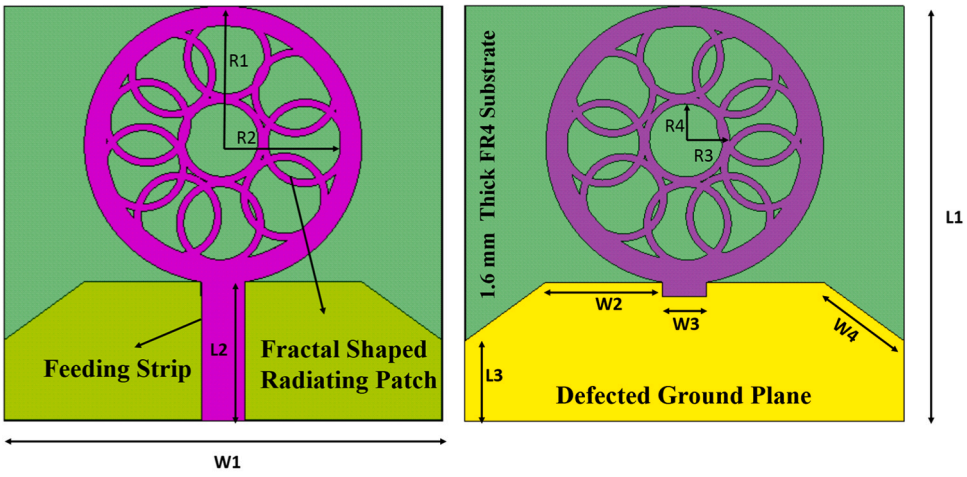


Fig. 1. (a) Front view (b) Back view.

the antenna types, patch antennas hold significance owing to their compact size, ease of fabrication, and cost-effectiveness. The various antenna types with distinct frequency responses, such as helical antennas [14–18] and monopole antennas, have been proposed to achieve broad bandwidth and ultra-wideband characteristics. In [19], compact two-port and four-port circularly polarized antennas were developed for MIMO operation to facilitate cognitive communication. A CPW-fed pentagonal fractal antenna was designed for near-field microwave imaging applications, particularly for breast cancer detection, as described in [20]. Furthermore, [21] discusses an ultra-wideband (UWB) fractal antenna based on a novel epsilon-shaped geometry for microwave applications in the medical field. [22] presents the design and fabrication of an innovative circular shaped fractal antenna in the ultra-wideband frequency for the WiMAX, WLAN, C and X Band Applications. The textile antenna was designed [23] by using hybrid Sierpinski-Minkowski fractal geometry for medical applications.

Numerous concepts for microstrip patch antennas have been proposed to achieve wide bandwidth. However, these antennas often entail complexity in fabrication and are relatively large in size. Therefore, achieving a broad frequency range while minimizing antenna size remains crucial. Particularly for breast cancer detection applications, smaller antenna sizes are preferable. Various miniaturization techniques were discussed in [24] and the use of metamaterial miniaturization technique was used in [25,26]. The technique of modifications in the ground plane was discussed in [27,28] for the breast cancer detection application.

The Major contribution of the research work involves

- Introducing the miniaturization technique-reshaping the patch, the fractal antenna.
- Design a compact fractal antenna achieving a return loss of less than − 10 dB for breast cancer detection purpose.
- Achieving the return loss over a wide frequency range preferably in UWB frequency range.

2. Design evolution of the proposed antenna

Table 1 shows the design evolutions of the planned antenna. In the iteration 1, the radiating antenna was configured as a circle with a 19 mm radius, featuring a feed line and a uniform ground. However, the accompanying graph distinctly illustrates that the reflection coefficient falls within the − 10 dB bandwidth only between 8 and 8.5 GHz, which is notably insufficient for breast cancer detection applications. Due to the low return loss observed in the antenna design of Iteration I, an antenna for Iteration II was developed by removing a substantial portion of the radiating element, resulting in a uniform ground plane antenna. However, the graph indicates a further decrease in reflection coefficients, ranging from 0 to − 3 dB, with inadequate resonance in the ultra-wideband frequency range. Consequently, the antenna was deemed unsuitable for breast cancer applications.

The antennas in Iterations I and II exhibited extremely inadequate return loss. Consequently, in the iteration III, modifications were implemented by introducing concentric circles of appropriate radius to form a fractal antenna, aiming to enhance radiation properties. Small annular rings with radius R3 and R4 were formed within the empty space of the main radiator, alongside annular rings with radius R1 and R2. However, this structure demonstrated the capability to generate a single resonance band only at 8 GHz. Additionally, the reflection coefficient remained very poor, fluctuating between 0 and 6 dB across most frequencies, rendering it unsuitable for breast cancer applications.

The removal or addition of a fractal structure to the radiating element did not significantly enhance the radiation properties of the antenna. Therefore, a modification was attempted on the ground plane. In Iteration IV, the antenna from Iteration II was utilized, and the ground plane was altered by shortening its length and introducing small triangular slots at the two outer edges. In Iteration V, the fractal antenna from Iteration III was employed with a rectangular ground plane to achieve the desired reflection coefficient. However, both of these iteration structures necessitate further improvement in the reflection coefficient to render the antenna suitable for breast cancer detection purposes. Therefore, a design incorporating two triangular slots and a rectangular slot at the inner top edge of the ground plane, along with a fractal-shaped radiator, was constructed. This antenna achieved ultra-wideband operation spanning from 3.1 to 10.5 GHz, with a − 10 dB impedance bandwidth covering the 2:1 VSWR range, rendering it suitable for breast cancer detection purposes. The detailed geometry is shown.

3. Final antenna design and geometry

Fig. 1 illustrates the overall structure of the proposed antenna (a). The initial step involves printing a concentric circle with radius R1 and R2 on the front side of a 1.6 mm thick FR4 substrate. The feeder line, with dimensions L1 × W1, feeds this radiating concentric circle. On the rear side of the substrate, the ground plane formation entails three processes. Firstly, a rectangular shape with dimensions L × W2 mm^2 is cut to create the ground plane. Secondly, a tiny rectangular slit, measuring L2 × W3 mm^2, is cut at the top center of the ground plane.

Table 2
Detailed dimension of the proposed broad-band antenna.

R_1	R_2	L	L_1	L_2	L_3	W	W_1	W_2	W_3	W_4
19	16	28.5	9.5	1	4	30	3	9.5	3	5

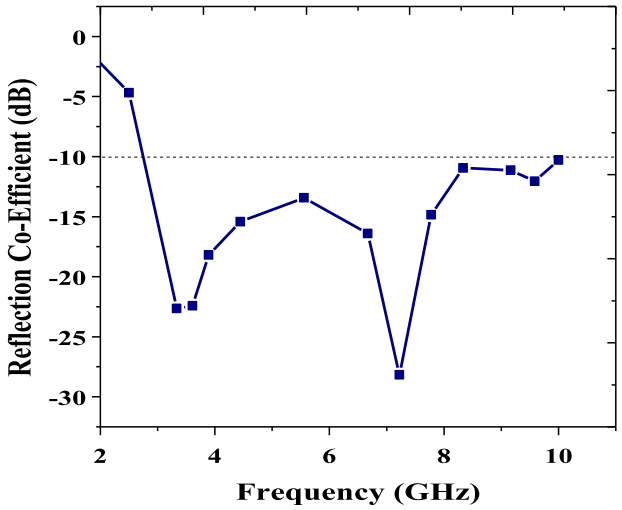


Fig. 2. Simulated reflection co-efficient of the proposed antenna.

Finally, as depicted in Fig. 1, two triangular-shaped slots with dimensions L3 × W4 are etched out of the ground plane at two side corners (b). The entire antenna is printed on a substrate with dimensions L × W mm^2. Table 2 provides the precise design specifications of the proposed antenna.

In Fig. 2, the simulated S-parameter of the proposed antenna is illustrated. The antenna covers a frequency spectrum from 3 to 10 GHz over a 10 dB impedance bandwidth, aligning with the ultra-wideband range recommended by the FCC.

4. Results and discussions

Fig. 3(a)–(c) depict the surface current distribution of the antenna at 4 GHz, 6 GHz, and 8 GHz, respectively. Across all three scenarios, the current distribution is highest near the edges of the concentric circle. This observation underscores the significance of the length of the concentric radiating circle in determining the frequency of operation within the range of 3–10.2 GHz.

Figs. 4(a) and 4 (b) illustrate the E-field distribution of the antenna. It is evident that the highest E-field distribution occurs near the upper edge of the radiating element, precisely opposite the peak current distribution. This observation indicates that the designed antenna closely adheres to boundary conditions.

Fig. 5(a) and (b) shows the three-dimensional radiation pattern of the proposed wideband antenna at 4 GHz and 6 GHz, respectively. It is evident that the antenna exhibits distinctive radiation characteristics throughout its operational band, achieving a maximum gain of 3.05 dBi,

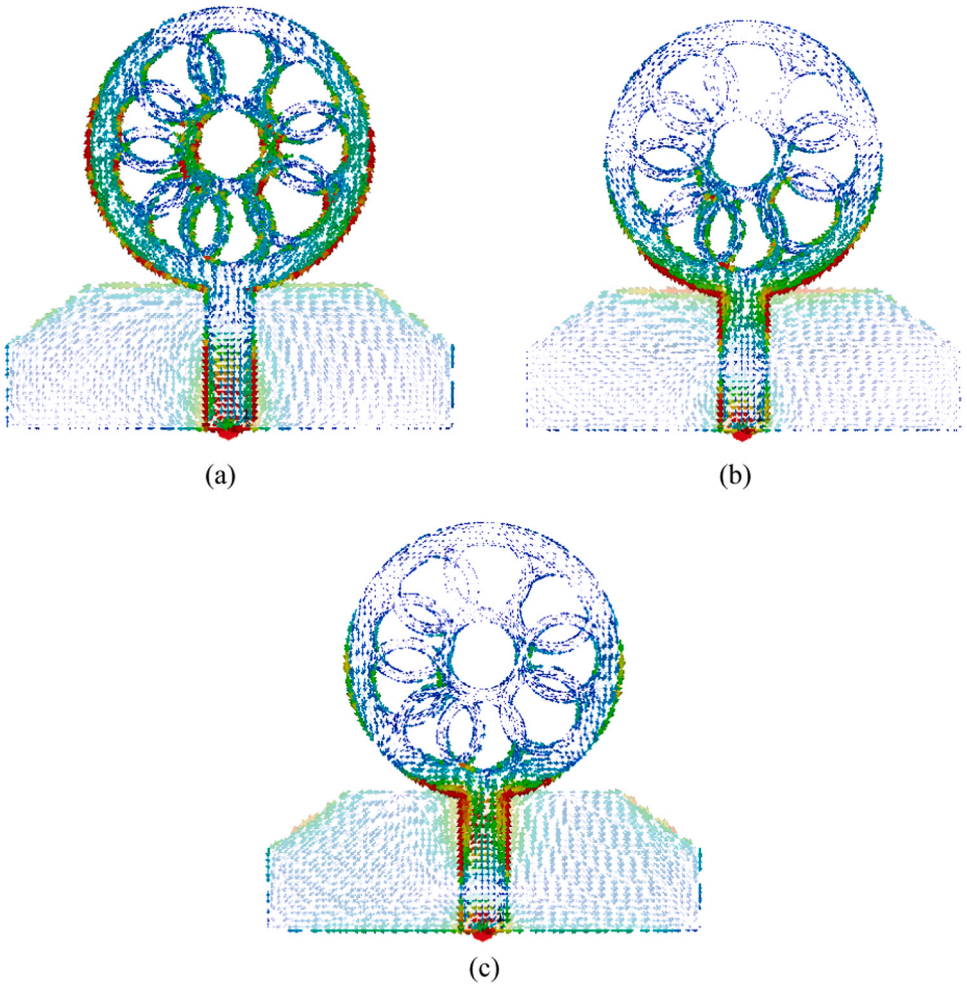


Fig. 3. Surface current distribution of the proposed antenna at (a) 4 GHz, (b) 6 GHz, (c) 8 GHz.

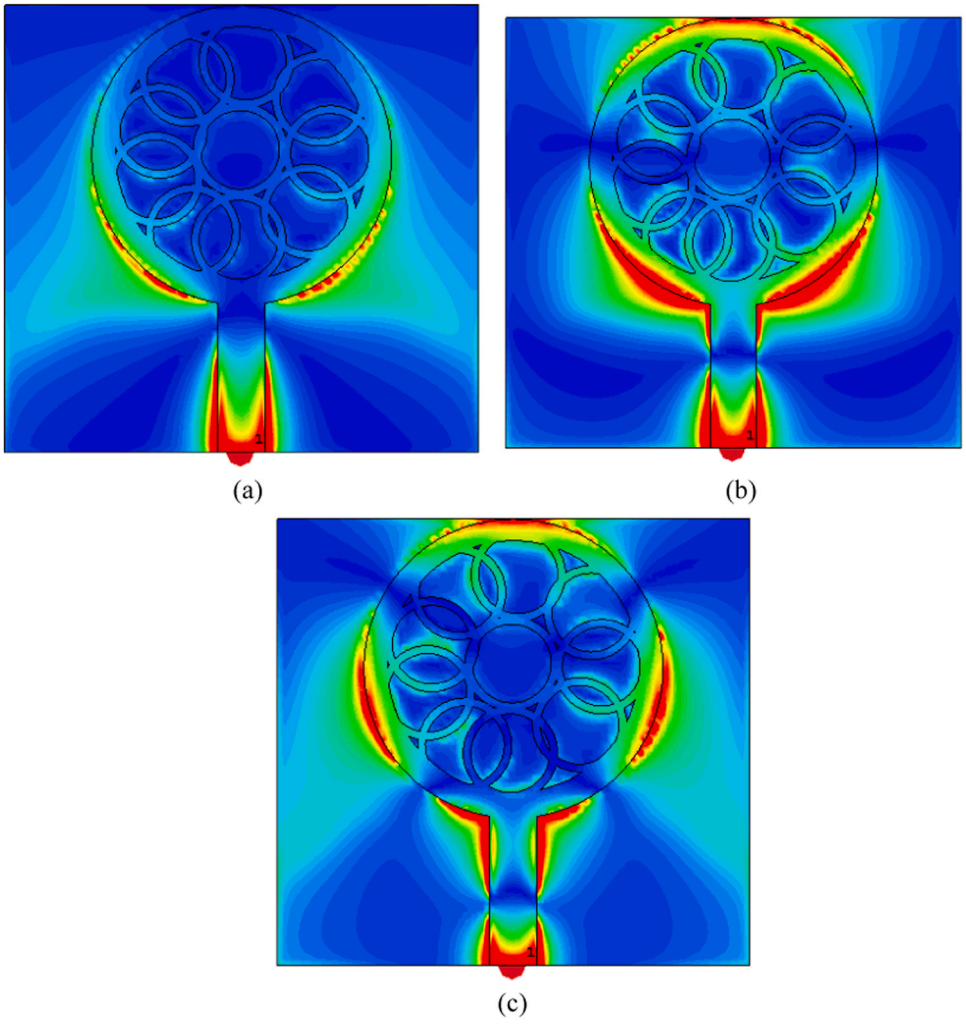


Fig. 4. Electric field distribution of the proposed antenna at (a) 4 GHz, (b) 6 GHz, (c) 8 GHz.

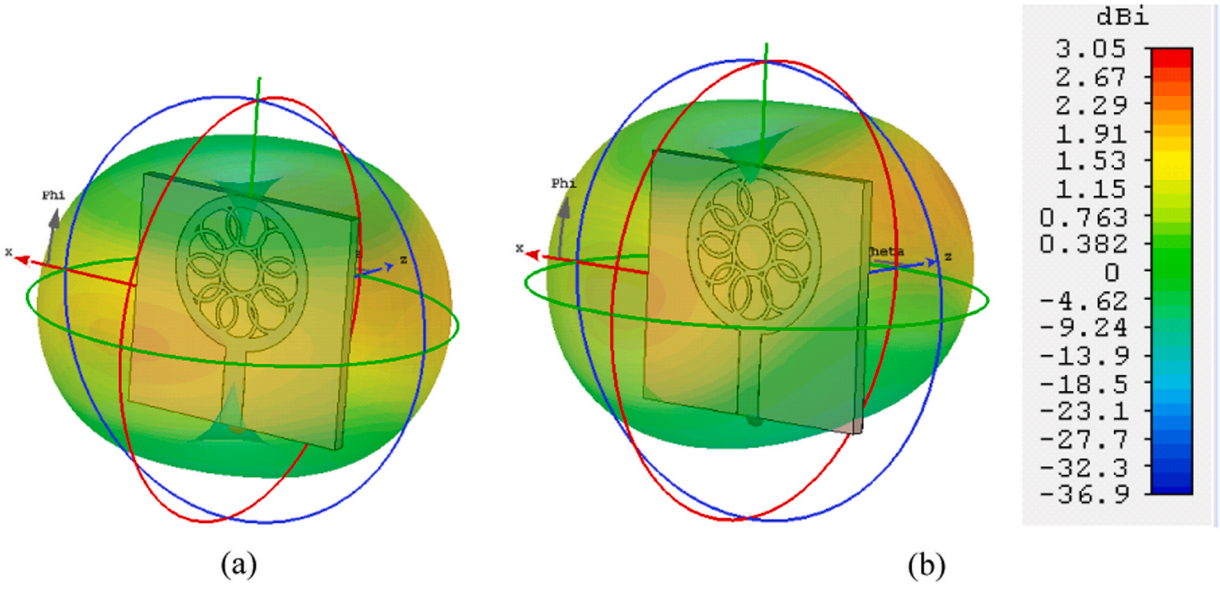


Fig. 5. Three dimensional radiation pattern of the antenna at (a) 4 GHz (b) 6 GHz.

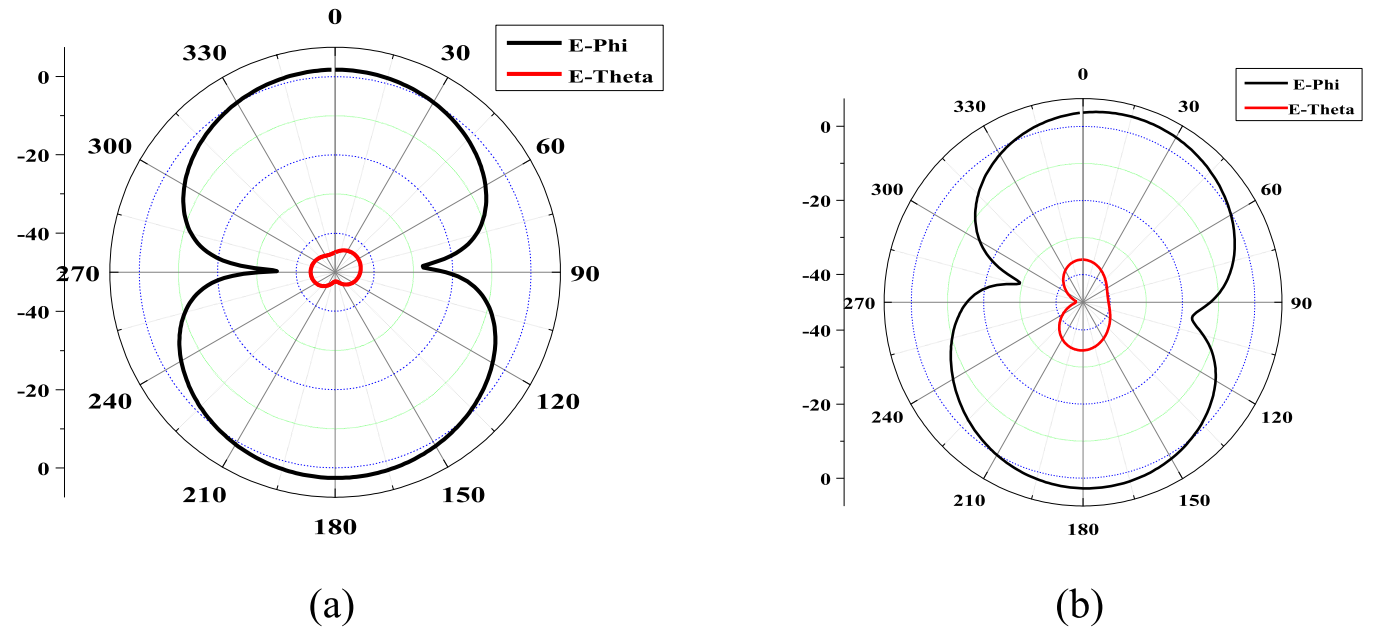


Fig. 6. 2D radiation pattern of the antenna at a) 4 GHz, (b) 6 GHz.

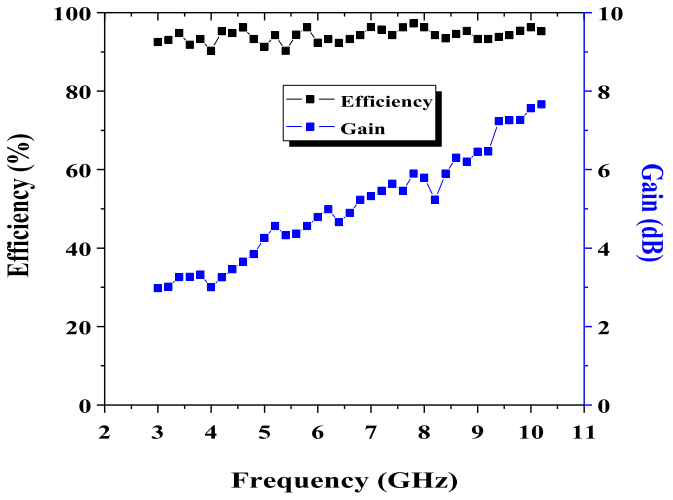


Fig. 7. Efficiency and gain of the ultra wide-band antenna.

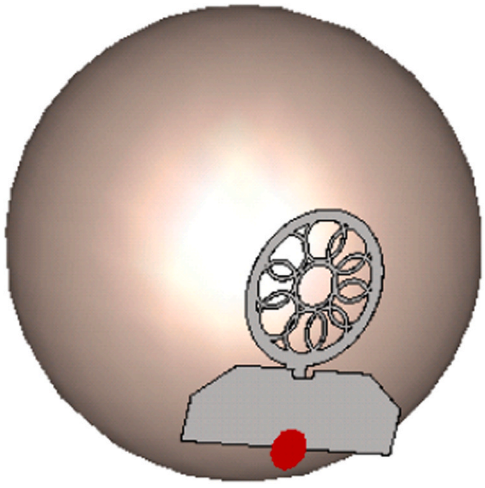


Fig. 8. Proposed antenna with breast Phantom.

Table 3
SAR of the proposed antenna at various frequency ranges.

Frequency (GHz)	3	5	7	9	10
SAR (W/Kg) for 1 g tissue	1.02	1.12	1.157	1.23	1.123
SAR (W/Kg) for 10 g tissue	0.986	0.975	0.998	0.978	0.985

crucial for UWB applications. Additionally, the antenna demonstrates a dipole-like radiation pattern within the desired operational band.

Fig. 6(a) and (b) depict the two-dimensional radiation pattern of the designed antenna at 4 and 6 GHz, respectively. It is evident that the antenna exhibits an omnidirectional radiation pattern without any significant nulls present throughout the pattern. Moreover, the cross-polarization is observed to be as low as − 40 dB.

The antenna maintains an overall efficiency of approximately 91–97 % across its operational bandwidth, which holds significant implications for breast cancer detection application. Additionally, it achieves substantial gains ranging from 3.24 to 7.9 dBi (Fig. 7).



Fig. 9. Simulated and measured return loss of the fractal antenna.

Table 3 displays the Specific Absorption Rate (SAR) of the proposed antenna across various Ultra-wide-band frequency ranges: 3 GHz, 5 GHz, 7 GHz, 9 GHz, and 10 GHz. SAR measurements at these

Table 4
Performance comparison of the proposed antenna.

Ref.	Dimension	Substrate	Dielectric constant	Frequency (GHz)
[14]	260 × 360	FR4	4.4	1–3
[15]	50 × 62 × 1.52	Taconic RF-35	3.5	1.69–8.61
[16]	50 × 50	FR4	4.4	2.9–12
[17]	50.8 × 62	FR4	4.4	1.3–20
[18]	32 × 35	FR4	4.4	3.1 – 10.6
[19]	44 × 22	FR4	4.4	2.65–3.02
[20]	24 × 30	Rogers RT/ Duroid 5880	2.2	3–12.7
[21]	26 × 22	FR4	4.4	2.5–20.95
[22]	40 × 34	FR4	4.4	6.0–9.82
[23]	53 × 36	Denim Jeans	1.6	2.08–6
Proposed Work	**28.5 × 30**	**FR4**	**4.4**	**3.2–10.6**

frequencies were conducted concerning both 1 g tissue and 10 g tissue models Fig 8. The SAR values of the proposed antenna design reach a maximum of 1.23 W/kg for 1 g tissue and 0.998 W/kg for 10 g tissue, both of which fall below the permissible limit of 1.6 W/kg. Hence, the proposed antenna design proves to be highly suitable for breast cancer detection, especially considering its testing proximity to the human body.

The antenna was fabricated using FR4 substrate and tested using Anritsu MS2036C VNA + Spectrum Analyzer for return loss. The Fig. 9 shows the simulated and measured return loss of the antenna and it is evident that the fabricated antenna works better than the simulation.

Table 4 presents a performance comparison of the proposed antenna with other works involving fractal antennas. While the dimensions of the antennas in literature [20,21] are smaller in comparison to the proposed fractal antenna, it is important to note that both antennas were designed for medical purposes, not specifically for breast cancer detection. The proposed antenna exhibits a gain ranging from 3.24 to 7.9 dBi within the frequency range of 3.2–10.6 GHz. In contrast, the gain of the antennas in [20,21] is 3.6 dBi at 8 GHz and 7.21 dBi at 20.95 GHz, respectively. Additionally, the efficiency of the proposed antenna is approximately 91–97 %, whereas the efficiency of the antennas in [20,21,23] is 80 %, 85 %, and 86 % respectively. The analysis suggests that the designed antenna offers superior radiation performance with smaller dimensions.

5. Conclusions

A compact Defected Ground Structure-based ultra-wideband fractal antenna tailored for breast cancer detection applications has been successfully developed, exhibiting outstanding radiation characteristics. The proposed antenna effectively covers the entire frequency range pertinent to UWB applications (3–10.2 GHz) while maintaining a 2:1 Voltage Standing Wave Ratio (VSWR). Across the spectrum, the antenna achieves an overall radiation efficiency of up to 92 % with a realized gain of 3.2 dBi. Additionally, the antenna exhibits a uniform radiation pattern within the desired operational band, distinguishing it from others of its kind. Notably, the maximum Specific Absorption Rate (SAR) values of the proposed antenna are 0.998 W/kg and 1.23 W/kg for 10 g and 1 g tissue, respectively. The study aims to develop a compact antenna for breast cancer detection but no clinical trials have been conducted. Future plans involve conducting trials and collaborating with healthcare institutions and experts to validate the effectiveness of UWB antenna-based breast cancer detection.

CRediT authorship contribution statement

Sadasivam S: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft. **Thulasi Bai V:** Supervision, Writing – review & editing, Project administration.

Declaration of Competing Interest

THERE ARE NO CONFLICT OF INTEREST IN THIS ARTICLE.

References

[1] N. Howlader, A.M. Noone, M. Krapcho, D. Miller, K. Bishop, S.F. Altekruse, C. L. Kosary, M. Yu, J. Ruhl, Z. Tatalovich, A. Mariotto, D.R. Lewis, H.S. Chen, E. J. Feuer, K.A. Cronin, SEER Cancer Statistics Review, National Cancer Institute, Bethesda, MD, USA, 2016, pp. 1975–2013 ([Online]. Available: ⟨http://seer.cancer.gov/archive/csr/1975_2013/⟩).

[2] R. Siegel, D. Naishadham, A. Jemal, Cancer statistics, 2013, CA, A Cancer J. Clin., vol. 63(no. 1), 2013, pp. 11–30.

[3] C. Li, Breast Cancer Epidemiology, Springer, New York, NY, USA, 2010.

[4] D. Wörtge, J. Moll, V. Krozer, B. Bazrafshan, F. Hübner, C. Park, T. Vogl, Comparison of X-ray-Mammography and planar UWB microwave imaging of the breast: first results from a patient study, Diagnostics 8 (3) (2018) 54.

[5] S. Nass, I. Henderson, J. Lashof, Mammography and Beyond: Developing Technologies for the Early Detection of Breast Cancer, Nat. Cancer Policy Board, Inst. Med., Commission Life Stud, Nat. Res. Council, Washington, DC, USA, 2001.

[6] P.T. Huynh, A.M. Jarolimek, S. Daye, The false-negative mammogram. RadioGraphics 18 (5) (1998) 1137–1154.

[7] P.J. Kornguth, F.J. Keefe, K.R. Wright, D.M. Delong, Mammography pain in women treated conservatively for breast cancer, J. Pain 1 (4) (2000) 268–274.

[8] N. Nikolova, Microwave imaging for breast cancer, IEEE Microw. Mag. 12 (7) (2011) 78–94.

[9] L.L. Humphrey, M. Helfand, B.K. Chan, S.H. Woolf, Breast cancer screening: a summary of the evidence for the US preventive services task force, Ann. Intern. Med. 137 (5) (2002) 347–360.

[10] R.D. Rosenberg, W.C. Hunt, M.R. Williamson, F.D. Gilliland, P.W. Wiest, C. A. Kelsey, C.R. Key, M.N. Linver, Effects of age, breast density, ethnicity, and estrogen replacement therapy on screening mammographic sensitivity and cancer stage at diagnosis: review of 183,134 screening mammograms in Albuquerque, new Mexico, Radiology 209 (2) (1998) 511–518.

[11] V.P. Jackson, R.E. Hendrick, S.A. Feig, D.B. Kopans, Imaging of the radiographically dense breast, Radiology 188 (2) (1993) 297–301.

[12] S. Kwon, S. Lee, Recent advances in microwave imaging for breast cancer detection, Int. J. Biomed. Imag. 2016 (2016) 1–26.

[13] M.A. Stuchly, E.C. Fear, Microwave detection of breast cancer, IEEE Trans. Microw. Theory Technol. 48 (11) (2000) 1854–1863.

[14] Z. Wang, H. Zhang, Improvements in a high gain UWB antenna with corrugated edges, Prog. Electromagn. Res. C 6 (2009) 159–166.

[15] M. Abbak, M.N. Akıncı, M. Çayören, I. Akduman, Experimental microwave imaging with a novel corrugated Vivaldi antenna, IEEE Trans. Antennas Propag. 65 (6) (2017) 3302–3307.

[16] G.K. Pandey, M.K. Meshram, A printed high gain UWB vivaldi antenna design using tapered corrugation and grating elements, Int. J. RF Microw. Comput.-Aided Eng. 25 (7) (2015) 610–618.

[17] B. Biswas, R. Ghatak, D.R. Poddar, A fern fractal leaf inspired wideband antipodal Vivaldi antenna for microwave imaging system, IEEE Trans. Antennas Propag. 65 (11) (2017) 6126–6129.

[18] M.H. Bah, J. Hong, D.A. Jamro, Ground slotted monopole antenna design for microwave breast cancer detection based on time reversal music, Prog. Electromagn. Res. C 59 (4) (2015) 117–126.

[19] H. Islam, S. Das, T. Ali, T. Bose, S. Dhar, B. Rai, P. Kumar, Compact circularly polarized 2 and 4 port multiple input multiple output antennas with bandstop filter isolation technique, Alex. Eng. J. 66 (2023) 357–376.

[20] M.A. Khan, U. Rafique, H.Ş. Savci, A.N. Nordin, S.H. Kiani, S.M. Abbas, Ultra-wideband pentagonal fractal antenna with stable radiation characteristics for microwave imaging applications, Electronics 11 (13) (2022) 2061.

[21] H.T. Sediq, J. Nourinia, C. Ghobadi, B. Mohammadi, A novel shaped ultrawideband fractal antenna for medical purposes, Biomed. Signal Process. Control 80 (2023) 104363.

[22] M. Marzouk, Y. Rhazi, I.H. Nejdi, F.E. Zerrad, M. Saih, S. Ahmad, M. Hussein, et al., Ultra-wideband compact fractal antenna for WiMAX, WLAN, C and X band applications, Sensors 23 (9) (2023) 4254.

[23] Harvinder Kaur, Chawla Paras, Design and evaluation of a fractal wearable textile antenna for medical applications, Wirel. Pers. Commun. 128 (1) (2023) 683–699.

[24] Muhammad Umar Khan, Said Sharawi Mohammad, Mittra Raj, Microstrip patch antenna miniaturisation techniques: a review, IET Microw. Antennas Propag. 9 (9) (2015) 913–922.

[25] S. Sadasivam, Bai Thulasi, Karthick Ramanathan S, A.K. Sheik Mohammed, Rishwanth M. S, Sivakami B, Design of Rectangular shape Meta-Material Antenna for C band applications, in: 2021 Emerging Trends in Industry 4.0 (ETI 4.0), IEEE, 2021, pp. 1–4.

[26] M. Vinoth, R. Vallikannu, Design and analysis of metamaterial patch antenna 5G and X band applications, in: Proceedings of the 2021 International Conference on Computer Communication and Informatics (ICCCI), IEEE, 2021, pp. 1–6.
[27] S. Sadasivam, V. Thulasi Bai, A compact diamond shaped ultra-wide band antenna system for diagnosing breast cancer, Technol. Health Care (2023) 57–68.
[28] Sarmad Nozad Mahmood, et al., Full ground ultra-wideband wearable textile antenna for breast cancer and wireless body area network applications, Micromachines 12 (3) (2021) 322.